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Question 1 [5]

1. Which traversal always visits the keys of a BST in ascending sorted order?
a) Pre-order b) In-order c) Post-order d) Level-order
2. Which of the following correctly defines a **Full Binary Tree**?
a) Every node has either exactly 2 children or 0 children (leaf)
b) Every level is completely filled
c) Every leaf is at the same depth
d) Every node must have at least 1 child
3. Keys 50, 30, 70, 20, 40, 60, 80, 10 are inserted into a BST in this order. How many comparisons are needed to reach key = 10 starting from the root? (a single node has height 0)
a) 2 b) 3 c) 4 d) 5
4. When deleting a BST node that has two children, it is typically replaced by:
a) Its parent
b) Its in-order successor or in-order predecessor
c) The root node
d) Any random leaf node
5. In a Full Binary Tree, if the number of **internal nodes** (nodes with 2 children) is i , how many **leaf nodes** does the tree have?
a) i b) $i + 1$ c) $2i$ d) $i - 1$

Question 2 [6]

You are given the root of a BST, along with two integers, low and high. Write a function `rangeSumBST()` that returns the sum of all node values in the BST that lie between low and high(inclusive).

No extra data structures are allowed.

Assume the Node class has: (int) val, (Node) left, (Node) right

Example:

BST:

```
      50
     /  \
    30   70
   / \  / \
  20 40 60 80
```

`rangeSumBST(root, 30, 65)`

→ $30 + 40 + 50 + 60 = 180$

Python:

```
def rangeSumBST(root, low, high):  
    # TODO
```

Java:

```
static int rangeSumBST(Node root, int  
low, int high) {  
  
    // TODO
```

Question 3 [4]

Insert the following keys into an empty BST, in this order: 45, 25, 65, 15, 35, 55, 75, 5, 20

- Show, step by step, how the tree looks after each key is inserted (or clearly draw the final resulting tree).
- Now delete the node 25 (which has two children) using its **in-order successor**, and show the tree after the deletion.
- What is the height of the final tree? (assume a single node has height 0)